

Class 10: Structural Bioinformatics 1

Mikaela Joya (PID: A17295169)

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Background

The main repository of high-resolution structural data on biomolecules is called the **Protein Data Bank** (PDB).

PDB statistics

What is in the PDB in terms of molecule type and structure determination method?

Read a CSV file of current PDB stats obtained from <https://www.rcsb.org/stats/summary>

```
pdb <- read.csv("Data Export Summary.csv")
pdb
```

	Molecular.Type	X.ray	EM	NMR	Integrative	Multiple.methods
1	Protein (only)	180,758	23,111	12,813	348	229
2	Protein/Oligosaccharide	10,488	3,741	34	8	11
3	Protein/NA	9,205	6,751	287	26	8
4	Nucleic acid (only)	3,154	250	1,578	3	15
5	Other	178	27	35	4	0

6	Oligosaccharide (only)	11	0	6	0	1
	Neutron Other Total					
1	84 32	217,375				
2	1 0	14,283				
3	0 0	16,277				
4	3 1	5,004				
5	0 0	244				
6	0 4	22				

Q1. What percentage of structures in the PDB are solved by X-Ray and Electron Microscopy.

```
pdb$X.ray
```

```
[1] "180,758" "10,488" "9,205" "3,154" "178" "11"
```

This print out above `pdb$X.ray` is “character” not “numeric”. Therefore I can’t do math with it. We need to fix this...

The two functions that help here are `sub()` and `as.numeric`

```
# We want to get rid (or sub out) commas:
x <- pdb$X.ray
tmp <- sub(",", "" , x)
sum( as.numeric(tmp))
```

```
[1] 203794
```

We could make a function to do this:

```
rm.comma <- function(x) {
  tmp <- sub(",", "" , x)
  sum( as.numeric(tmp))
}
```

```
n.total <- rm.comma (pdb$Total)
n.xray <- rm.comma(pdb$X.ray)
n.em <- rm.comma(pdb$EM)

n.xray/n.total * 100
```

```
[1] 80.48577
```

```
n.em/n.total * 100
```

```
[1] 13.38046
```

~80.5% of PDB structures are solved by X-ray crystallography and ~13.4% are solved by Electron Microscopy.

We could also use a different import function for this CSV that speaks American (i.e. deals with commas in numbers in a comma separated value file)

```
library(readr)

pdb <- read_csv("Data Export Summary.csv")
```

```
Rows: 6 Columns: 9
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (1): Molecular Type
```

```
dbl (4): Integrative, Multiple methods, Neutron, Other
```

```
num (4): X-ray, EM, NMR, Total
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
n.tot <- sum(pdb$Total)
n.xray <- sum(pdb$`X.ray`)
```

```
Warning: Unknown or uninitialised column: `X.ray`.
```

```
n.em <- sum(pdb$EM)
```

```
n.xray/n.tot * 100
```

```
[1] 0
```

```
n.em/n.tot * 100
```

```
[1] 13.38046
```

Q. How many total protein structures are there in the data set?

```
pdb$Total[1]
```

```
[1] 217375
```

The total number of protein sequences in the dataset is 202,556,314.

```
217375/202556314 * 100
```

```
[1] 0.1073158
```

Key-point: We have a very, very small structural coverage of known proteins (~0.1%). Most structures we know about (~80%) come from one method (X-ray crystallography).

Q2. What proportion of structures in the PDB are protein?

```
pdb[, 2:ncol(pdb)] <- lapply(pdb[, 2:ncol(pdb)], function(x) {  
  as.numeric(gsub(",", "", x))  
})  
  
n.tot <- sum(pdb[, 2:ncol(pdb)], na.rm = TRUE)  
  
protein <- sum(  
  pdb[pdb$`Molecular Type` == "Protein (only)", 2:ncol(pdb)],  
  na.rm = TRUE  
)  
  
protein / n.tot * 100
```

```
[1] 85.84941
```

There are ~85.8% of structures in the PDB that are protein only structures.

Q3. Type HIV in the PDB website search box on the home page and determine how many HIV-1 protease structures are in the current PDB?

There are 1,227 HIV-1 protease structures in the current PDB.

Visualizing the HIV-1 protease structure

Main stand alone web version with all features is at <https://molstar.org/viewer/>.

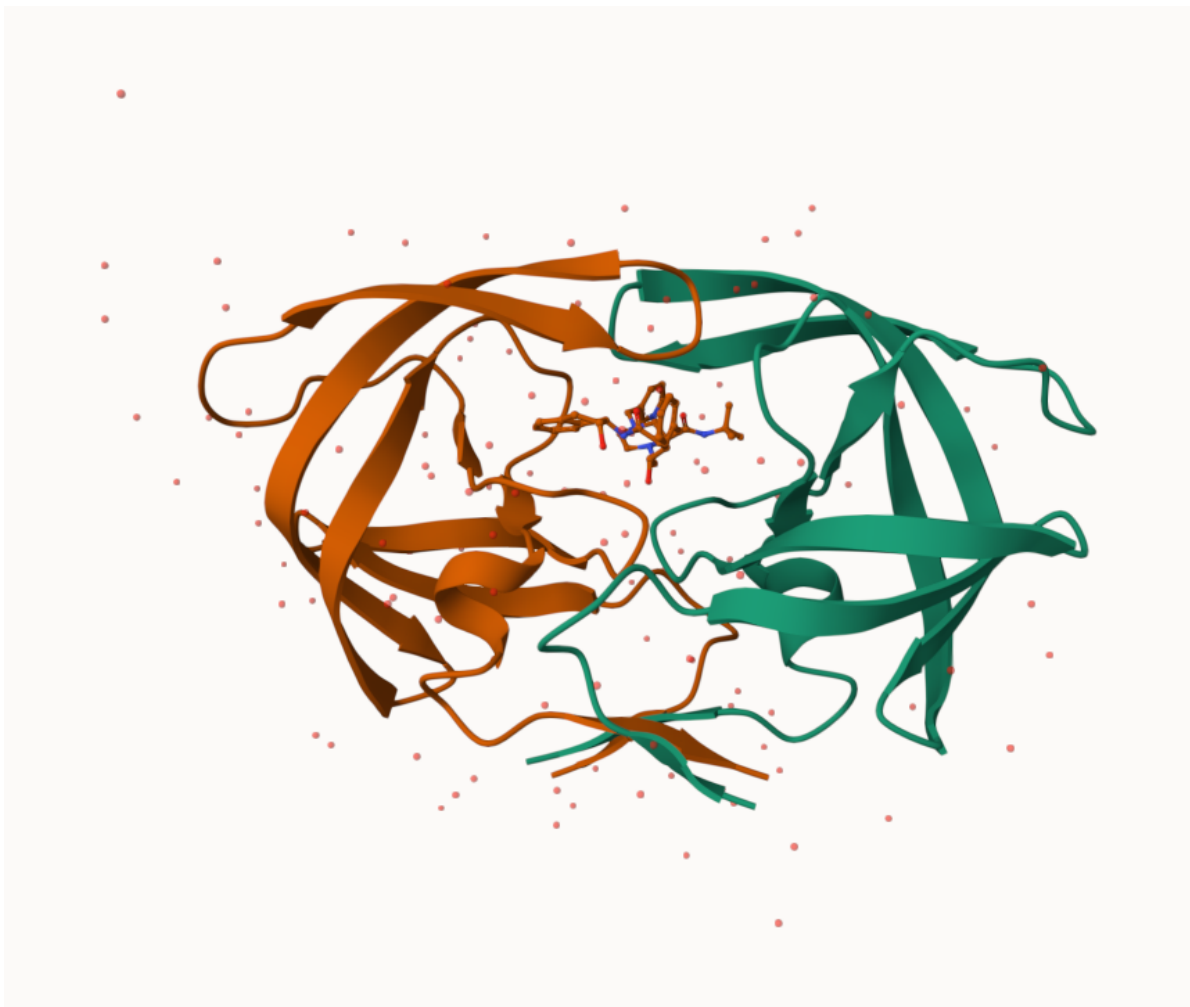


Figure 1: The HIV-protease enzyme is a homodimer with two chains

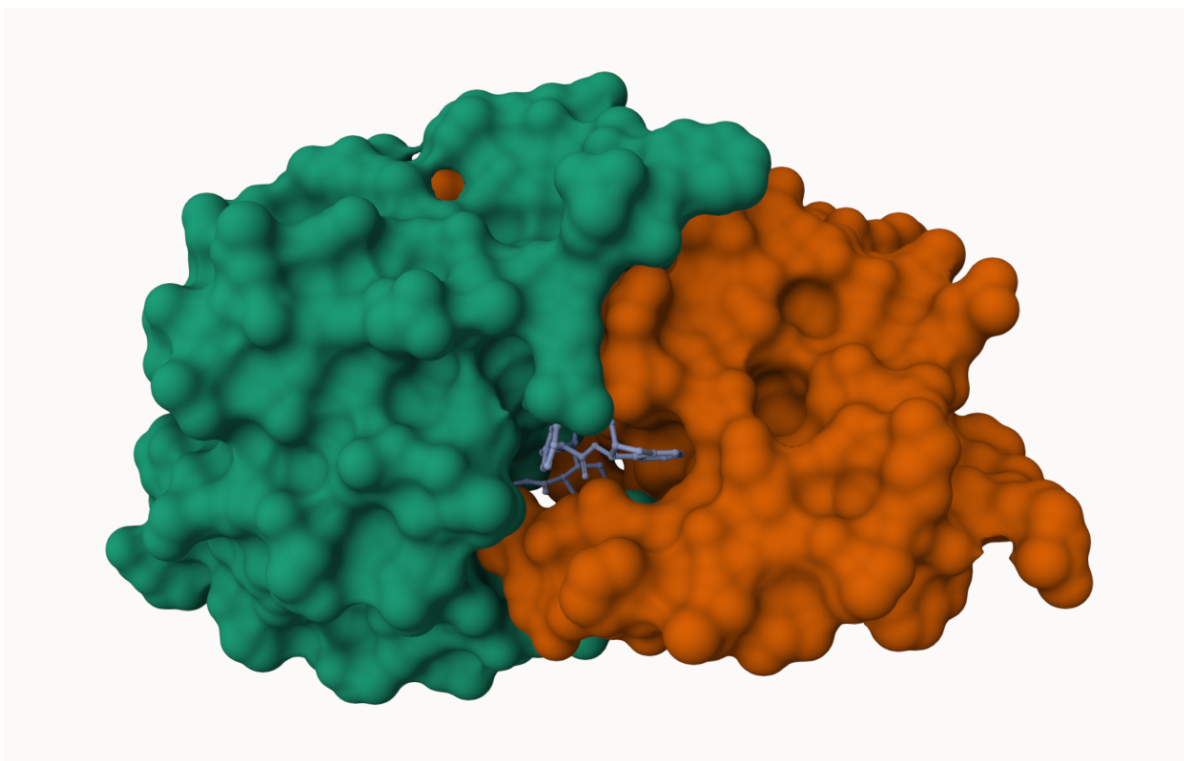


Figure 2: Surface display showing the binding cleft site for the inhibitor (drug) molecule

Q4. Water molecules normally have 3 atoms. Why do we see just one atom per water molecule in this structure?

There is only one atom per water molecule because the X-ray structures show only the oxygen atom of water. The hydrogen atoms are quite small and scatter the X-rays less so they are not that visible in the PDB structures.

Q5. There is a critical “conserved” water molecule in the binding site. Can you identify this water molecule? What residue number does this water molecule have?

The critical “conserved” water molecule in the binding site is the water that connects the inhibitor and the residues flapping (specifically HOH 301).

Now you should be able to produce an image similar or even superior to Figure 2 and save it to an image file.

Q6. Generate and save a figure clearly showing the two distinct chains of HIV-protease along with the ligand. You might also consider showing the catalytic residues ASP 25 in each chain and the critical water (we recommend “Ball & Stick” for these side-chains). Add this figure to your Quarto document.

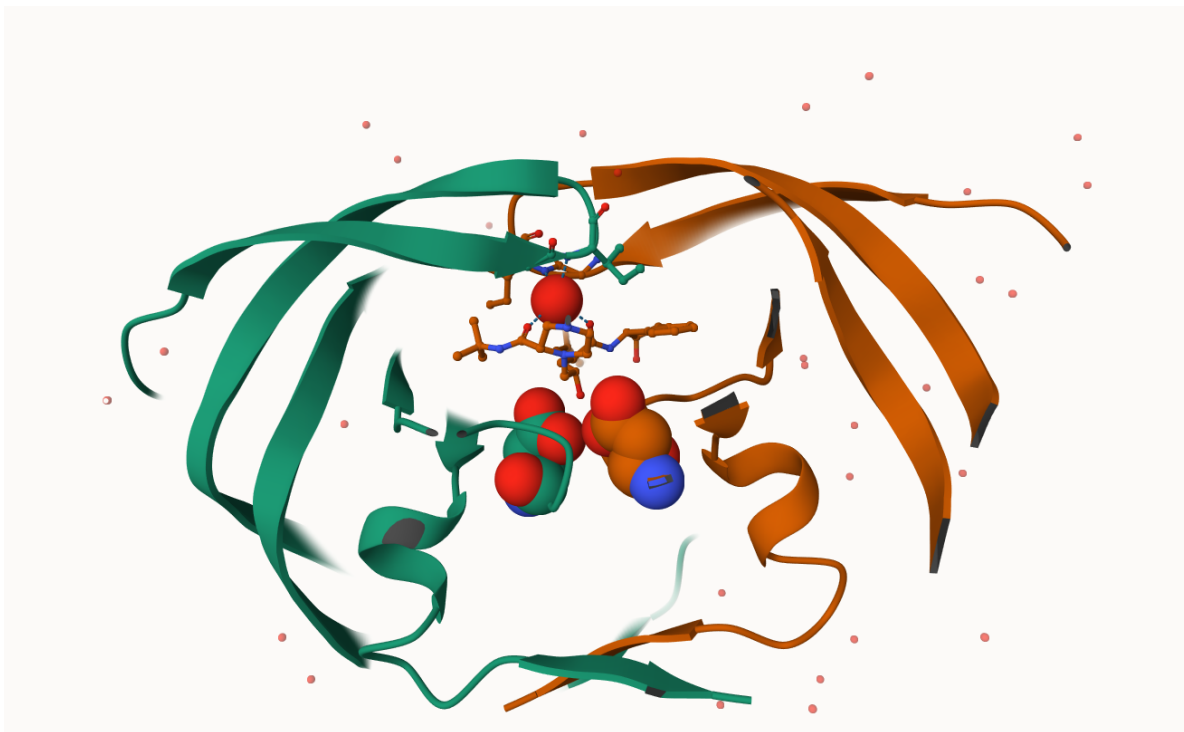


Figure 3: Spacefill display of a catalytic ASP25 amino acids and key binding site watermolecule

Discussion Topic: Can you think of a way in which indinavir, or even larger ligands and substrates, could enter the binding site? Large ligands and substrates could enter the binding site through the flexible flap regions where they can open and close allowing the ligand and substrates in before closing.

Getting started with the Bio3D package

Bio3D is an R package from CRAN for structural bioinformatics

```
library(bio3d)

pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

pdb

```
Call: read.pdb(file = "1hsg")
```

```
Total Models#: 1
```

```
Total Atoms#: 1686, XYZs#: 5058 Chains#: 2 (values: A B)
```

```
Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 172 (residues: 128)
```

```
Non-protein/nucleic resid values: [ HOH (127), MK1 (1) ]
```

```
Protein sequence:
```

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

head(pdb\$atom)

	type	eleno	elety	alt	resid	chain	resno	insert	x	y	z	o	b
1	ATOM	1	N	<NA>	PRO	A	1	<NA>	29.361	39.686	5.862	1	38.10
2	ATOM	2	CA	<NA>	PRO	A	1	<NA>	30.307	38.663	5.319	1	40.62
3	ATOM	3	C	<NA>	PRO	A	1	<NA>	29.760	38.071	4.022	1	42.64
4	ATOM	4	O	<NA>	PRO	A	1	<NA>	28.600	38.302	3.676	1	43.40
5	ATOM	5	CB	<NA>	PRO	A	1	<NA>	30.508	37.541	6.342	1	37.87
6	ATOM	6	CG	<NA>	PRO	A	1	<NA>	29.296	37.591	7.162	1	38.40
	segid	elesy	charge										
1	<NA>	N	<NA>										
2	<NA>	C	<NA>										
3	<NA>	C	<NA>										
4	<NA>	O	<NA>										
5	<NA>	C	<NA>										
6	<NA>	C	<NA>										

There are lots of functions that can work with these `pdb` objects


```
head( pdbseq( pdb ) )
```

```
  1    2    3    4    5    6  
"P" "Q" "I" "T" "L" "W"
```

Q7. How many amino acid residues are there in this `pdb` object?

There are 198 residues.

Q8. Name one of the two non-protein residues?

One of the non-protein residues is HOH (127). The other one is MK1 (1).

Q9. How many protein chains are in this structure?

There are 2 protein chains: chain A and B.

We can have a quick interactive view of any of these `pdb` objects:

```
library(bio3dview)  
  
# view.pdb(pdb)
```

Let's try a custom view

```
#view.pdb(pdb, colorScheme="sse")
```

Q. Create a custom view highlighting the active site ASP (`resno=25`), the two chains (in your choice of colors) and the ligand all on a custom color background

```
library(NGLVieweR)  
  
active.site <- atom.select(pdb,resno=25)  
  
#view.pdb(pdb,  
  #cols = c("red","blue"),  
  #highlight = active.site,  
  #highlight.style = "spacefill",  
  #backgroundColor= "pink") |>  
#setRock()
```

Predict the flexibility of a given structure

Let's do a Normal Mode Analysis (NMA) to predict the flexibility of a given `pdb` object:

```
adk <- read.pdb("6s36")
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, `rm.alt=TRUE`

A quick structure summary

```
adk
```

```
Call: read.pdb(file = "6s36")
```

```
Total Models#: 1
  Total Atoms#: 1898, XYZs#: 5694 Chains#: 1 (values: A)

  Protein Atoms#: 1654 (residues/Calpha atoms#: 214)
  Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)

  Non-protein/nucleic Atoms#: 244 (residues: 244)
  Non-protein/nucleic resid values: [ CL (3), HOH (238), MG (2), NA (1) ]
```

Protein sequence:

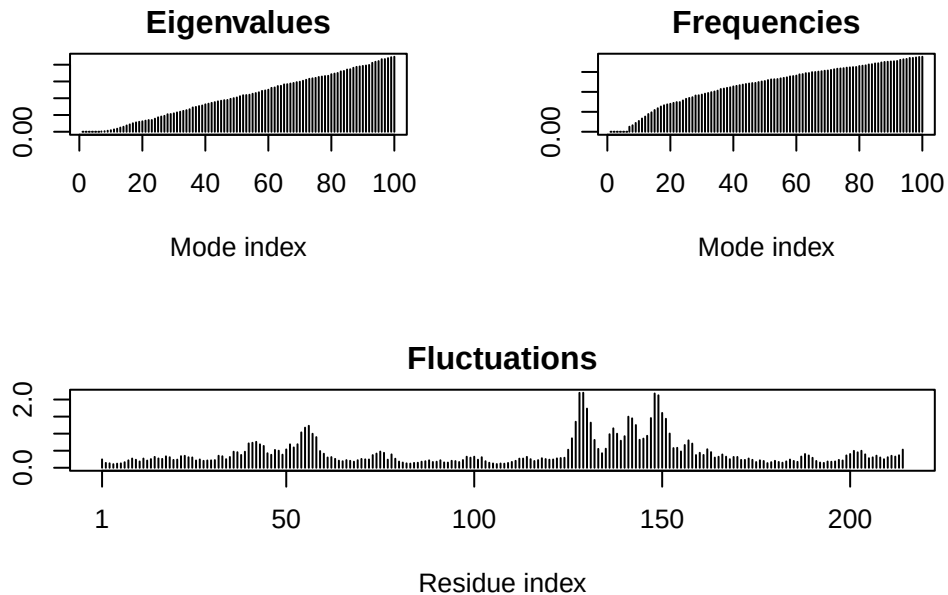
```
MRILLGAPGAGKGTQAQFIMEKYGIPQISTGDMRLRAAVKSGSELGKQAKDIMDAGKLV
DELVIALVKERIAQEDCRNGFLLDGFPRTPQADAMKEAGINVDYVLEFDVPDELIVDKI
VGRRVHAPSGRVYHVKFNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
```

```
+ attr: atom, xyz, seqres, helix, sheet,
      calpha, remark, call
```

```
m <- nma( adk )
```

```
Building Hessian...      Done in 0.012 seconds.
Diagonalizing Hessian... Done in 0.27 seconds.
```

```
plot(m)
```



View the results

```
#view.nma(m)
```

Write out the results for viewing in Mol-star:

```
mktrj(m, file="nma.pdb")
```

Comparative analysis of the ADK family

Q10. Which of the packages above is found only on BioConductor and not CRAN?

msa is the package only found in Bioconductor not CRAN.

Q11. Which of the above packages is not found on BioConductor or CRAN?

Bio3Dview is the package not found in Bioconductor nor CRAN because it's installed straight from GitHub.

Q12. True or False? Functions from the pak package can be used to install packages from GitHub and BitBucket?

True.

Our first step is find a sequence for this family. We will use the database ID “lake_A” here:

```
library(bio3d)
id <- "lake_A"

aa <- get.seq(id)
```

Warning in get.seq(id): Removing existing file: seqs.fasta

Fetching... Please wait. Done.

aa

```

      1      .      .      .      .      .      .      60
pdb|1AKE|A  MRIILLGAPGAGKGTQAQFIMEKYGIPQISTGDMMLRAAVKSGSELGKQAKDIMDAGKLV
      1      .      .      .      .      .      .      60

      61      .      .      .      .      .      .      120
pdb|1AKE|A  DELVIALVKERIAQEDCRNGFLLDGFPRPTIPQADAMKEAGINVDYVLEFDVPDELIVDRI
      61      .      .      .      .      .      .      120

      121      .      .      .      .      .      .      180
pdb|1AKE|A  VGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKDDQEETVRKRLVEYHQMTAPLIG
      121      .      .      .      .      .      .      180

      181      .      .      .      214
pdb|1AKE|A  YYSKEAEAGNTKYAKVDGTPVAEVRADLEKILG
      181      .      .      .      214
```

Call:

```
read.fasta(file = outfile)
```

Class:

```
fasta
```

Alignment dimensions:

```
1 sequence rows; 214 position columns (214 non-gap, 0 gap)
```

+ attr: id, ali, call

Q13. How many amino acids are in this sequence, i.e. how long is this sequence?

There are 214 amino acids in the sequence

Search for related sequences

```
blast <- blast.pdb(aa)
```

Searching ... please wait (updates every 5 seconds) RID = 00B8T2GS014

.....

Reporting 96 hits

```
head( blast$hit.tbl)
```

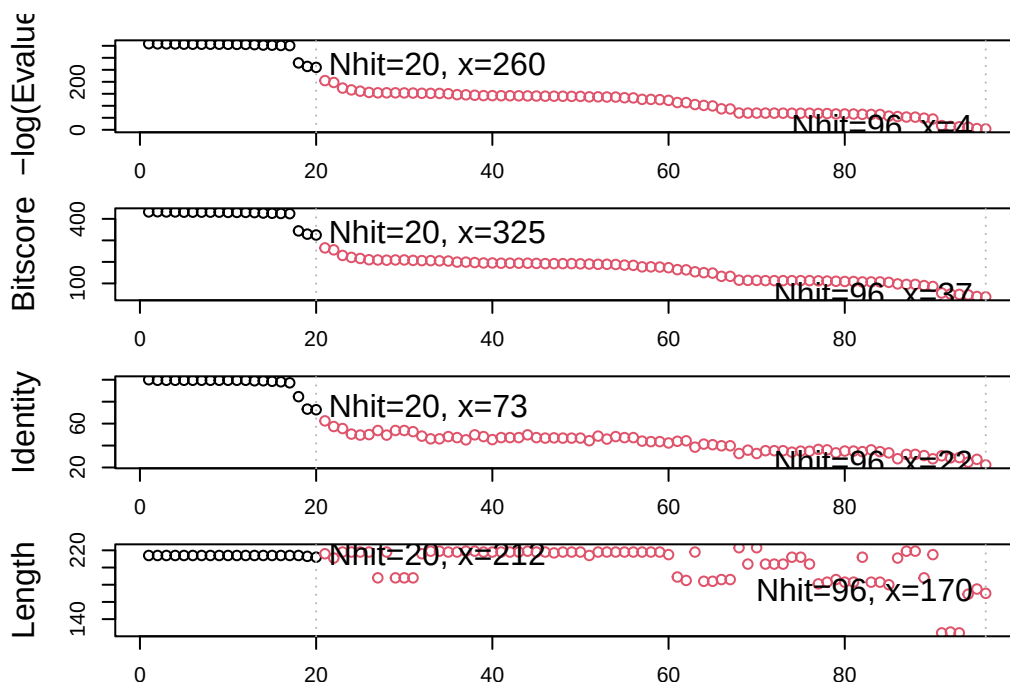
	queryid	subjectids	identity	alignmentlength	mismatches	gapopens	q.start
1	Query_4024661	1AKE_A	100.000	214	0	0	1
2	Query_4024661	8BQF_A	99.533	214	1	0	1
3	Query_4024661	4X8M_A	99.533	214	1	0	1
4	Query_4024661	6S36_A	99.533	214	1	0	1
5	Query_4024661	9R6U_A	99.533	214	1	0	1
6	Query_4024661	9R71_A	99.533	214	1	0	1

	q.end	s.start	s.end	evaluate	bitscore	positives	mlog.evalue	pdb.id	acc
1	214	1	214	1.82e-156	432	100.00	358.6044	1AKE_A	1AKE_A
2	214	21	234	2.97e-156	433	100.00	358.1147	8BQF_A	8BQF_A
3	214	1	214	3.25e-156	432	100.00	358.0246	4X8M_A	4X8M_A
4	214	1	214	4.77e-156	432	100.00	357.6409	6S36_A	6S36_A
5	214	1	214	1.06e-155	431	99.53	356.8424	9R6U_A	9R6U_A
6	214	1	214	1.25e-155	431	99.53	356.6775	9R71_A	9R71_A

```
hits <- plot(blast)
```

* Possible cutoff values: 260 3
Yielding Nhits: 20 96

* Chosen cutoff value of: 260
Yielding Nhits: 20



```
hits$pdb.id
```

```
[1] "1AKE_A" "8BQF_A" "4X8M_A" "6S36_A" "9R6U_A" "9R71_A" "8Q2B_A" "8RJ9_A"
[9] "6RZE_A" "4X8H_A" "3HPR_A" "1E4V_A" "5EJE_A" "1E4Y_A" "3X2S_A" "6HAP_A"
[17] "6HAM_A" "8PVW_A" "4K46_A" "4NP6_A"
```

```
files <- get.pdb(hits$pdb.id, path="pdb", split=TRUE, gzip=TRUE)
```

```
Warning in get.pdb(hits$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/1AKE.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/8BQF.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/4X8M.pdb.gz exists. Skipping download
```

```
Warning in get.pdb(hits$pdb.id, path = "pdb", split = TRUE, gzip = TRUE):
pdb/6S36.pdb.gz exists. Skipping download
```

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R6U.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/9R71.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8Q2B.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8RJ9.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6RZE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4X8H.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3HPR.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4V.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/5EJE.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/1E4Y.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/3X2S.pdb.gz exists. Skipping download

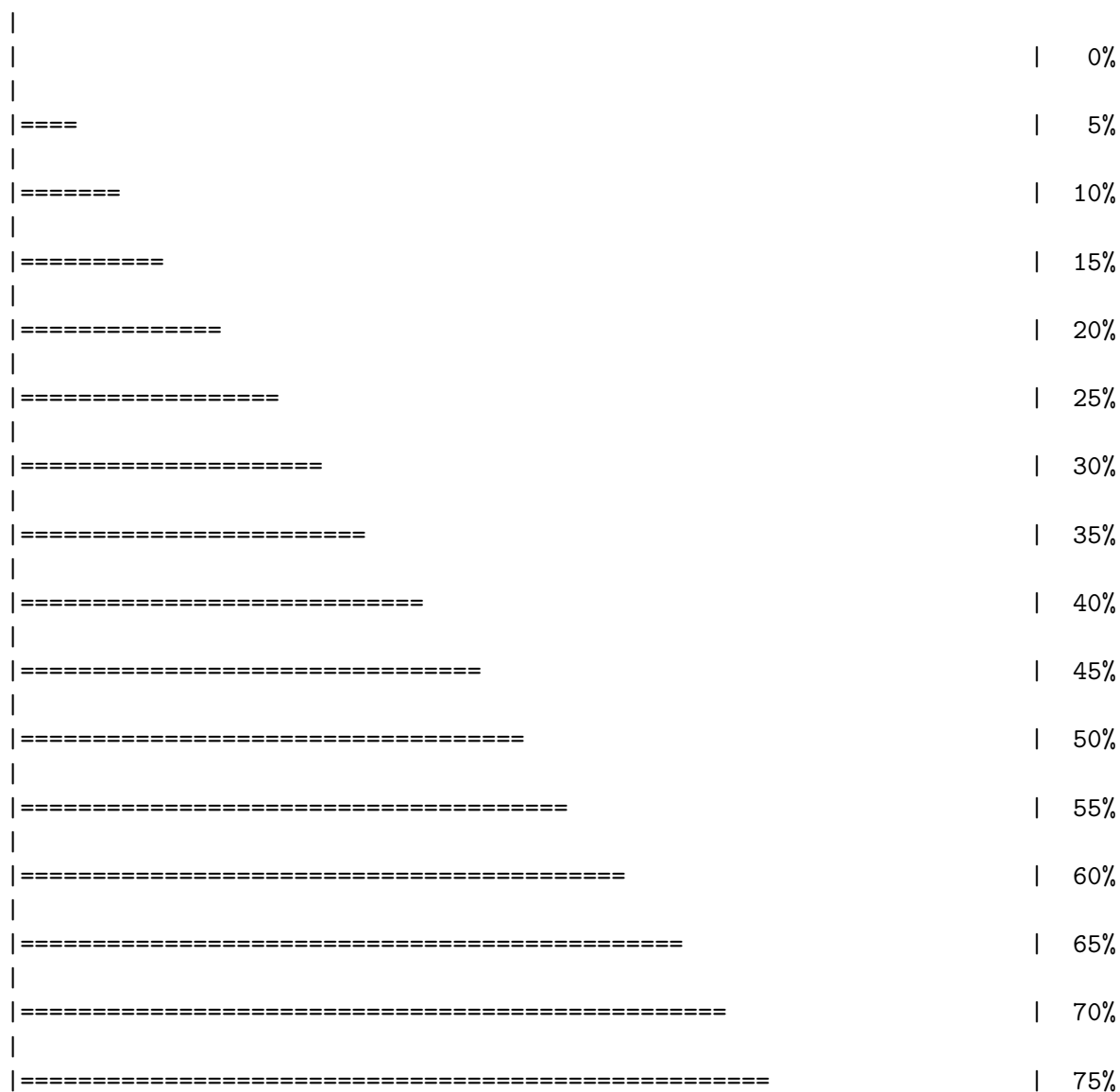
Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAP.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/6HAM.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/8PVW.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4K46.pdb.gz exists. Skipping download

Warning in get.pdb(hits\$pdb.id, path = "pdbs", split = TRUE, gzip = TRUE):
pdbs/4NP6.pdb.gz exists. Skipping download




```

|
|=====| 80%
|
|=====| 85%
|
|=====| 90%
|
|=====| 95%
|
|=====| 100%

```

Align and superpose all these ADK structures

```
pdbbs <- pdbaln(files, fit = TRUE, exefile="msa")
```

Reading PDB files:

```

pdbbs/split_chain/1AKE_A.pdb
pdbbs/split_chain/8BQF_A.pdb
pdbbs/split_chain/4X8M_A.pdb
pdbbs/split_chain/6S36_A.pdb
pdbbs/split_chain/9R6U_A.pdb
pdbbs/split_chain/9R71_A.pdb
pdbbs/split_chain/8Q2B_A.pdb
pdbbs/split_chain/8RJ9_A.pdb
pdbbs/split_chain/6RZE_A.pdb
pdbbs/split_chain/4X8H_A.pdb
pdbbs/split_chain/3HPR_A.pdb
pdbbs/split_chain/1E4V_A.pdb
pdbbs/split_chain/5EJE_A.pdb
pdbbs/split_chain/1E4Y_A.pdb
pdbbs/split_chain/3X2S_A.pdb
pdbbs/split_chain/6HAP_A.pdb
pdbbs/split_chain/6HAM_A.pdb
pdbbs/split_chain/8PVW_A.pdb
pdbbs/split_chain/4K46_A.pdb
pdbbs/split_chain/4NP6_A.pdb

```

```

    PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE

```

```

.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
..  PDB has ALT records, taking A only, rm.alt=TRUE
.... PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
.   PDB has ALT records, taking A only, rm.alt=TRUE
..

```

Extracting sequences

```

pdb/seq: 1   name: pdbc/split_chain/1AKE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 2   name: pdbc/split_chain/8BQF_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 3   name: pdbc/split_chain/4X8M_A.pdb
pdb/seq: 4   name: pdbc/split_chain/6S36_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 5   name: pdbc/split_chain/9R6U_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 6   name: pdbc/split_chain/9R71_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 7   name: pdbc/split_chain/8Q2B_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 8   name: pdbc/split_chain/8RJ9_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 9   name: pdbc/split_chain/6RZE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 10  name: pdbc/split_chain/4X8H_A.pdb
pdb/seq: 11  name: pdbc/split_chain/3HPR_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 12  name: pdbc/split_chain/1E4V_A.pdb
pdb/seq: 13  name: pdbc/split_chain/5EJE_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 14  name: pdbc/split_chain/1E4Y_A.pdb
pdb/seq: 15  name: pdbc/split_chain/3X2S_A.pdb
pdb/seq: 16  name: pdbc/split_chain/6HAP_A.pdb
pdb/seq: 17  name: pdbc/split_chain/6HAM_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 18  name: pdbc/split_chain/8PVW_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE
pdb/seq: 19  name: pdbc/split_chain/4K46_A.pdb
           PDB has ALT records, taking A only, rm.alt=TRUE

```

pdb/seq: 20 name: pdbs/split_chain/4NP6_A.pdb

pdbs

	1	.	.	.	40
[Truncated_Name:1] 1AKE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:2] 8BQF_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:3] 4X8M_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:4] 6S36_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:5] 9R6U_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:6] 9R71_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:7] 8Q2B_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:8] 8RJ9_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:9] 6RZE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:10] 4X8H_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:11] 3HPR_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:12] 1E4V_A.pdb	--MRIILLGAPVAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:13] 5EJE_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:14] 1E4Y_A.pdb	--MRIILLGALVAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:15] 3X2S_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:16] 6HAP_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:17] 6HAM_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:18] 8PVW_A.pdb	--MRIILLGAPGAGKGTQAQFIMEKYGIPQISTG	DMLRAA			
[Truncated_Name:19] 4K46_A.pdb	--MRIILLGAPGAGKGTQAQFIMAKFGIPQISTG	DMLRAA			
[Truncated_Name:20] 4NP6_A.pdb	NAMRIILLGAPGAGKGTQAQFIMEKFGIPQISTG	DMLRAA			
	*****	*****	*~*****		
	1	.	.	.	40
	41	.	.	.	80
[Truncated_Name:1] 1AKE_A.pdb	VKSGSELGKQAKDIMDAGKLV	TDELVIALVKERIAQEDCR			
[Truncated_Name:2] 8BQF_A.pdb	VKSGSELGKQAKDIMDAGKLV	TDELVIALVKERIAQE---			
[Truncated_Name:3] 4X8M_A.pdb	VKSGSELGKQAKDIMDAGKLV	TDELVIALVKERIAQEDCR			
[Truncated_Name:4] 6S36_A.pdb	VKSGSELGKQAKDIMDAGKLV	TDELVIALVKERIAQEDCR			
[Truncated_Name:5] 9R6U_A.pdb	VKSGSELGAQAKDIMDAGKLV	TDELVIALVKERIAQEDCR			
[Truncated_Name:6] 9R71_A.pdb	VKSGSELGKQAKDIMDAGKLV	TDELVIALVKERIAQEDCR			
[Truncated_Name:7] 8Q2B_A.pdb	VKSGSELGKQAKDIMDAGKLV	TDELVIALVKERIAQEDCR			
[Truncated_Name:8] 8RJ9_A.pdb	VKSGSELGKQAKDIMDAGKLV	TDELVIALVKERIAQEDCR			
[Truncated_Name:9] 6RZE_A.pdb	VKSGSELGKQAKDIMDAGKLV	TDELVIALVKERIAQEDCR			
[Truncated_Name:10] 4X8H_A.pdb	VKSGSELGKQAKDIMDAGKLV	TDELVIALVKERIAQEDCR			
[Truncated_Name:11] 3HPR_A.pdb	VKSGSELGKQAKDIMDAGKLV	TDELVIALVKERIAQEDCR			
[Truncated_Name:12] 1E4V_A.pdb	VKSGSELGKQAKDIMDAGKLV	TDELVIALVKERIAQEDCR			
[Truncated_Name:13] 5EJE_A.pdb	VKSGSELGKQAKDIMDACKLV	TDELVIALVKERIAQEDCR			

[Truncated_Name:14] 1E4Y_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:15] 3X2S_A.pdb	VKSGSELGKQAKDIMDCGKLVDELVIALVKERIAQEDSR
[Truncated_Name:16] 6HAP_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVRERICQEDSR
[Truncated_Name:17] 6HAM_A.pdb	IKSGSELGKQAKDIMDAGKLVDEIIIALVKERICQEDSR
[Truncated_Name:18] 8PVW_A.pdb	VKSGSELGKQAKDIMDAGKLVDELVIALVKERIAQEDCR
[Truncated_Name:19] 4K46_A.pdb	IKAGTELGKQAKSVIDAGQLVSDDIILGLVKERIAQDDCA
[Truncated_Name:20] 4NP6_A.pdb	IKAGTELGKQAKAVIDAGQLVSDDIILGLIKERIAQADCE

```

^* *^*** *** ^~*   **^*~^~^~*~^*** *
41      .      .      .      80

```

[Truncated_Name:1] 1AKE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD	81	120
[Truncated_Name:2] 8BQF_A.pdb	-GFLLDGFPR TIPQADAMKEAGINVDYVIEFDVPDELIVD		
[Truncated_Name:3] 4X8M_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:4] 6S36_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:5] 9R6U_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:6] 9R71_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDALIVD		
[Truncated_Name:7] 8Q2B_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:8] 8RJ9_A.pdb	NGFLLAGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:9] 6RZE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:10] 4X8H_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:11] 3HPR_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:12] 1E4V_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:13] 5EJE_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:14] 1E4Y_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:15] 3X2S_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:16] 6HAP_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:17] 6HAM_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:18] 8PVW_A.pdb	NGFLLDGFPR TIPQADAMKEAGINVDYVLEFDVPDELIVD		
[Truncated_Name:19] 4K46_A.pdb	KGFLLDGFPR TIPQADGLKEVGVVVDYVIEFDVADSVIVE		
[Truncated_Name:20] 4NP6_A.pdb	KGFLLDGFPR TIPQADGLKEMGINVDYVIEFDVADDVIVE		

```

**** *****^~** *^ *****^***** * ^**^
81      .      .      .      120

```

[Truncated_Name:1] 1AKE_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD	121	160
[Truncated_Name:2] 8BQF_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD		
[Truncated_Name:3] 4X8M_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD		
[Truncated_Name:4] 6S36_A.pdb	KIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD		
[Truncated_Name:5] 9R6U_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD		
[Truncated_Name:6] 9R71_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD		
[Truncated_Name:7] 8Q2B_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKA		
[Truncated_Name:8] 8RJ9_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD		

[Truncated_Name:9] 6RZE_A.pdb	AIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:10] 4X8H_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:11] 3HPR_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDGTGEELTTRKD
[Truncated_Name:12] 1E4V_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:13] 5EJE_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:14] 1E4Y_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:15] 3X2S_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:16] 6HAP_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:17] 6HAM_A.pdb	RIVGRRVHAPSGRVYHVKNPPKVEGKDDVTGEELTTRKD
[Truncated_Name:18] 8PVW_A.pdb	RILKR--GETSGRV-----D
[Truncated_Name:19] 4K46_A.pdb	RMAGRRRAHLASGRTYHNVNPPKVEGKDDVTGEDLVIRED
[Truncated_Name:20] 4NP6_A.pdb	RMAGRRRAHLPSGRTYHVNVNPPKVEGKDDVTGEDLVIRED
	^ * ***
	121 . . . 160
	161 . . . 200
[Truncated_Name:1] 1AKE_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:2] 8BQF_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:3] 4X8M_A.pdb	DQEETVRKRLVEWHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:4] 6S36_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:5] 9R6U_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:6] 9R71_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:7] 8Q2B_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:8] 8RJ9_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:9] 6RZE_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:10] 4X8H_A.pdb	DQEETVRKRLVEYHQMTAALIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:11] 3HPR_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:12] 1E4V_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:13] 5EJE_A.pdb	DQEECVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:14] 1E4Y_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:15] 3X2S_A.pdb	DQEETVRKRLCEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:16] 6HAP_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:17] 6HAM_A.pdb	DQEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:18] 8PVW_A.pdb	DNEETVRKRLVEYHQMTAPLIGYYSKEAEAGNTKYAKVDG
[Truncated_Name:19] 4K46_A.pdb	DKEETVLARLGVYHNQTAPLIAYYGKEAEAGNTQYLKFDG
[Truncated_Name:20] 4NP6_A.pdb	DKEETVRARLNVYHTQTAPLIEYYGKEAAAGKTQYLKFDG
	* * * * * ^ * * * * * * * * * *
	161 . . . 200
	201 . 216
[Truncated_Name:1] 1AKE_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:2] 8BQF_A.pdb	TKPVAEVRADLEKIL-
[Truncated_Name:3] 4X8M_A.pdb	TKPVAEVRADLEKILG

[Truncated_Name:4] 6S36_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:5] 9R6U_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:6] 9R71_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:7] 8Q2B_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:8] 8RJ9_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:9] 6RZE_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:10] 4X8H_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:11] 3HPR_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:12] 1E4V_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:13] 5EJE_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:14] 1E4Y_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:15] 3X2S_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:16] 6HAP_A.pdb	TKPVCEVRADLEKILG
[Truncated_Name:17] 6HAM_A.pdb	TKPVCEVRADLEKILG
[Truncated_Name:18] 8PVW_A.pdb	TKPVAEVRADLEKILG
[Truncated_Name:19] 4K46_A.pdb	TKAVAEVSAELEKALA
[Truncated_Name:20] 4NP6_A.pdb	TKQVSEVSADIAKALA
	** * ** *~* *
201	. 216

Call:

```
pdbaln(files = files, fit = TRUE, exefile = "msa")
```

Class:

```
pdbs, fasta
```

Alignment dimensions:

```
20 sequence rows; 216 position columns (182 non-gap, 34 gap)
```

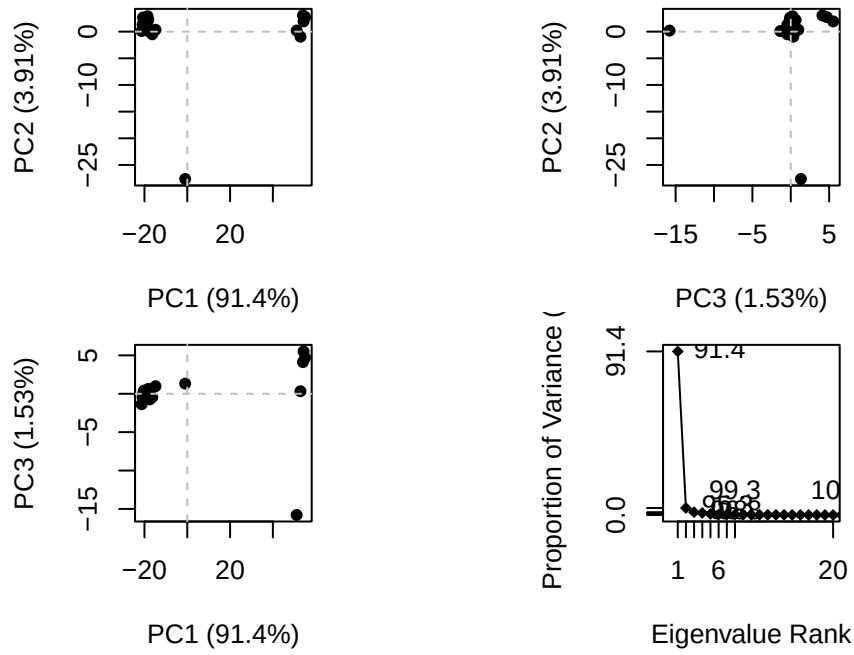
```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

Quick interactive structural view

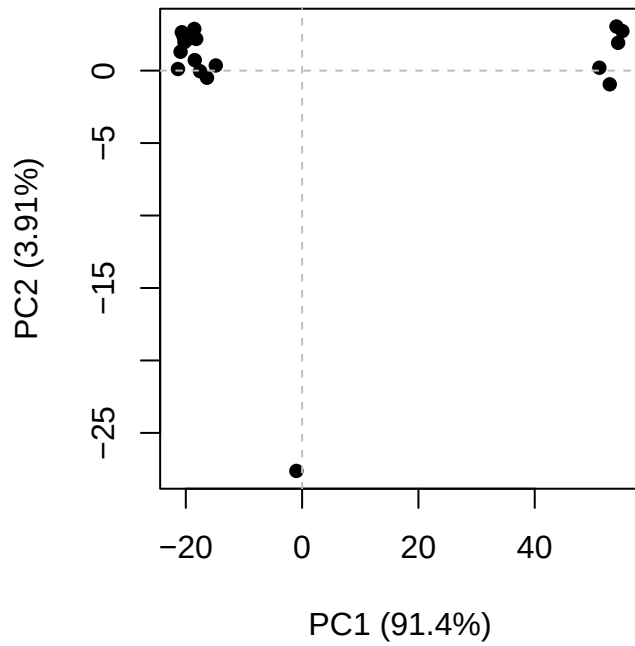
```
#view.pdbs(pdbs)
```

PCA of all this structural data (x, y, and z atom coordinates):

```
pc <- pca(pdbs)
plot(pc)
```



```
plot(pc, 1:2)
```



Interactive view of the PC1 captured structural differences:

```
#view.pca(pc)
```

```
mktrj(pc, file="pca.pdb")
```